

Unit 1 - Quantum Mechanics

1. State de-Broglie hypothesis of matter waves
2. Derive an expression for the de-Broglie wavelength of a particle
3. Obtain an expression for the de-Broglie wavelength of a charged particle
4. Derive an expression for the de-Broglie wavelength of an electron particle
5. State Heisenberg's Uncertainty Principle
6. Explain the broadening of spectral line using Heisenberg's Uncertainty Principle
7. Set up one dimensional time independent Schrodinger Wave equation
8. Discuss the physical significance of wave function (ψ)
9. Mention the properties of wave function (ψ) **or** What are well behaved wave functions
10. What are eigen wave functions? Mention their properties.
11. All numerical on de Broglie wavelength and Heisenberg uncertainty principle.
12. Apply Schrodinger wave equation for a particle (or free particle) confined in an one dimensional potential well of infinite depth (or particle in box of infinite depth or height) and obtain its energy eigen values
13. Normalize the wave function of particle in a box or
14. Graphically explain the probability amplitude, probability density and energy eigen values of a particle in an infinite well at ground state, first excited state and second excited state.
15. Discuss the Schrodinger wave equation for a free particle
16. Apply Schrodinger wave equation for a particle potential well of finite depth (or particle in box of finite depth or height) and obtain its energy eigen values
17. Write the two and three dimensional Schrodinger wave equation for a particle in an one dimensional potential well of infinite depth
18. Numerical

Unit-II Electrical conductivity & Dielectrics

1. Mention the assumption of Classical free electron theory or Drude model
2. Mention the failure or drawbacks of Classical free electron theory or Drude model
3. Discuss the assumption of quantum free electron theory or Drude-Sommerfeld model
4. Define Fermi Energy at 0K and Above 0K.
5. Define Fermi Factor/ Fermi function. Show its variation with temperature
6. Derive an expression for the electron concentration in metal at 0K.
7. **Numerical**